

MANNING AGDM, AB, Canada

WMO: 712290

Lat: 56.9739N

Lon: 117.4508W

Elev: 1499

StdP: 13.92

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-27.0	-41.1	0.5	-33.8	-33.6	0.9	-26.9	27.0	35.9	23.3	33.6	3.8	190	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	82.6	63.0	79.3	61.7	76.3	60.3	66.3	77.6	64.3	75.3	62.5	73.1	9.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	88.0	71.0	59.9	81.3	69.2	58.1	76.2	67.0	31.8	77.8	30.2	75.2	28.8	72.9	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	17.9	15.6	DB	-40.8	88.4	6.3	2.3	-45.3	90.1	-48.9	91.5	-52.4	92.8	-57.0	94.5	(4)
(5)			WB	-40.7	69.8	5.6	3.5	-44.7	72.4	-48.0	74.5	-51.1	76.4	-55.2	79.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.8	4.3	7.5	18.3	37.8	49.7	58.2	61.7	58.3	49.2	36.8	16.2	5.4
	DBStd	23.99	16.79	14.90	14.37	8.89	7.91	5.30	4.55	5.78	7.74	9.10	13.68	15.69
	HDD50	6993	1416	1191	982	374	102	5	0	6	105	416	1015	1381
	HDD65	11442	1881	1611	1447	817	476	211	123	216	475	874	1465	1846
	CDD50	1063	0	0	0	7	91	250	362	265	81	7	0	0
	CDD65	37	0	0	0	0	1	6	20	9	1	0	0	0
	CDH74	1191	0	0	0	9	94	234	462	308	81	3	0	0
	CDH80	224	0	0	0	1	16	37	88	67	15	0	0	0
	WSAvg	7.0	6.4	6.4	7.2	8.1	8.1	7.2	6.6	6.2	7.1	7.3	6.4	6.4
	PrecAvg	16.3	0.9	0.6	0.7	0.7	1.6	2.9	2.8	1.9	1.4	0.9	0.9	0.8
(15) Precipitation	PrecMax	21.3	2.2	1.4	2.0	1.9	3.6	5.3	5.1	4.9	4.1	2.4	2.1	2.8
	PrecMin	9.9	0.2	0.2	0.1	0.0	0.6	0.7	1.1	0.7	0.0	0.2	0.2	0.1
	PrecStd	3.6	0.6	0.3	0.4	0.4	0.7	1.4	1.0	1.1	0.8	0.5	0.6	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.8	42.9	50.2	73.8	82.2	84.5	86.4	86.2	82.2	71.1	51.4	42.3
		MCWB	37.8	36.7	41.1	51.6	57.7	62.0	67.0	64.7	61.7	54.0	43.1	37.3
	2%	DB	40.1	38.8	44.9	63.7	75.9	79.7	82.4	81.1	75.3	60.9	44.4	37.5
		MCWB	35.7	34.2	37.8	46.9	53.8	60.2	64.8	63.4	58.1	48.1	38.1	33.5
	5%	DB	36.3	34.1	41.7	58.1	72.3	76.3	79.1	77.0	69.5	55.8	39.6	33.7
		MCWB	33.3	31.1	35.9	44.3	52.1	58.9	63.9	61.5	55.0	45.3	35.1	30.6
	10%	DB	30.5	29.4	38.5	53.9	68.3	73.1	75.9	73.5	65.0	51.1	34.4	29.0
		MCWB	27.9	26.8	34.0	42.4	50.7	57.5	62.2	60.0	53.0	42.9	31.8	26.9
	0.4%	WB	38.3	37.4	41.7	52.4	59.3	66.2	71.1	68.0	62.9	55.2	43.5	37.7
		MCDB	43.0	41.8	49.2	70.5	77.2	76.9	81.3	81.4	77.8	69.9	50.5	42.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	34.8	38.4	48.2	55.8	62.8	67.4	65.3	59.6	49.3	38.6	33.9
		MCDB	39.7	38.5	43.8	61.5	71.9	74.6	78.6	76.6	71.8	58.6	43.5	37.2
	5%	WB	33.3	31.0	36.2	45.1	53.7	60.9	65.2	63.5	56.7	46.1	35.1	30.9
		MCDB	36.2	34.1	41.2	56.4	68.4	72.3	76.3	74.5	68.1	54.1	38.9	33.8
	10%	WB	27.9	26.7	34.1	42.9	51.7	59.0	63.3	61.4	53.8	43.6	32.0	26.7
		MCDB	30.5	29.4	38.2	52.7	65.1	70.0	73.6	71.6	62.8	50.3	34.4	29.1
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	21.5	22.9	22.2	27.4	24.8	25.3	26.2	25.7	20.0	17.4	17.5
		MCDBR	21.7	25.5	23.6	31.7	36.2	32.3	31.9	33.8	33.0	28.5	22.0	21.5
		MCWBR	18.2	21.6	17.6	17.9	17.7	16.1	17.1	17.8	17.3	17.2	16.7	18.1
	5% WB	MCDBR	21.4	24.9	22.4	29.6	31.8	27.4	28.9	30.9	32.7	26.5	21.6	21.3
		MCWBR	18.0	21.4	16.8	17.4	16.5	14.5	16.6	17.8	18.1	16.7	17.1	18.4
(40) Clear-Sky Solar Irradiance	taub		0.236	0.253	0.287	0.333	0.346	0.362	0.364	0.353	0.312	0.288	0.243	0.219
	taud		2.265	2.325	2.327	2.367	2.413	2.390	2.404	2.420	2.546	2.565	2.409	2.203
	Ebn at Noon		216	262	279	280	283	279	275	271	270	249	216	187
	Edh at Noon		15	22	29	33	34	35	34	31	23	18	13	12
(44) All-Sky Solar Radiation	RadAvg		204	507	1006	1490	1782	1875	1807	1471	1007	526	222	134
	RadStd		19	44	80	89	140	159	111	124	75	69	20	10

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (18 neighbors)	N/A	N/A	+3.31	N/A	N/A	N/A	N/A	N/A	+110	N/A	

Nomenclature: See separate page